

Extend calculator program given in assignment 3 with following functionalities:

- Identify INTEGER, DOUBLE and IDENTIFIER
- Assignment statement (marks1=20.5, marks2=10; total=marks1+marks2, etc.)
- Dynamic Symbol Table Management
- On each update save Symbol Table contents in a txt file (symbols.txt) and load while parser starts.
- Additional productions in the grammar

Grammar

The following context-free grammar can be used for arithmetic expressions:

statement $\rightarrow$ ID = expr

statement $\rightarrow$ expr

expr $\rightarrow$ expr + term

expr $\rightarrow$ expr - term

expr $\rightarrow$ term

term $\rightarrow$ term * power

term $\rightarrow$ term / power

term $\rightarrow$ power

power $\rightarrow$ factor ^ power

power $\rightarrow$ factor

factor $\rightarrow$ sin (expr)

factor $\rightarrow$ cos (expr)

factor $\rightarrow$ tan (expr)

factor $\rightarrow$ sqrt (expr)

factor $\rightarrow$ log (expr)

factor $\rightarrow$ (expr)

factor $\rightarrow$ INTEGER

factor $\rightarrow$ DOUBLE

factor $\rightarrow$ ID

Submission

Prepare a single document containing:

- Source code files
- Sample input
- Generated output